Using AI to Find (and Verify) Equilibrium

From Prisoner's Dilemma to a First-Price Procurement Auction

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The Core Principle

"Go back to the definition: would anyone want to deviate?"

Objective

1. Use AI as a calculation aid for finding Bayes-Nash Equilibrium
2. Develop the habit of verifying if AI's response actually satisfies the definition from economics.

Bayes-Nash Equilibrium

Definition

The strategy profile $s(\cdot)$ is a (pure-strategy) Bayes-Nash Equilibrium (BNE) if for every player $i \in I$ and type $\theta_i \in \Theta_i$,

$$s_i(\theta_i) \in \arg\max_{a_i} EU_i(a_i; s_{-i}).$$

All auction games we study in this class have a pure-strategy BNE.

A **strategy** $s_i(\cdot)$ is a *function* from types to actions. In equilibrium, **every type** of every player must be best-responding given others' strategies. This is what the $\forall\, \theta_i$ requires.

BNE vs. NE

- BNE applies in a **Bayesian game** (each player has a private type; rival's type is uncertain)
- Nash Equilibrium (NE) applies in a game with **no private information** (e.g. Prisoner's Dilemma)

$\Rightarrow$ NE is a special case of BNE with a single type.

The deviation principle is identical; what changes is that deviations in BNE must be checked type-by-type and rival uncertainty is **averaged out** via expectations.

Four Tools

1. **Check the math yourself**
2. **Ask a different AI to verify**
3. **Cross-check by re-asking** in a fresh chat

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1. **Check the math yourself**
2. **Ask a different AI to verify**
3. **Cross-check by re-asking** in a fresh chat
4. **Go back to the definition:** for every player and every type, confirm no profitable deviation exists

   $\Rightarrow$ Ask AI to execute this step!

Why Tool 4?

AI Knows What It Has Seen

First-price auction has been studied for decades → abundance of material for AI to learn from → will likely produce the correct equilibrium on the first try.

For a New Game, Nothing Is Guaranteed

For a *genuinely new* game or mechanism, AI has little or no relevant training data and can produce wrong answers
→ Tools 2 & 3 can make the same mistake.

This Is Not New

Theorists have always verified equilibria by searching for profitable deviations → We will execute that with AI

Prisoner's Dilemma: Game Setup

Two suspects, held separately. **Cooperate (C)** or **Defect (D)**.
Payoffs:

		Player 2	
		Cooperate	Defect
Player 1	Cooperate	$(-1, -1)$	$(-3,\ 0)$
	Defect	$(0, -3)$	$(-2, -2)$

No private information $\rightarrow$ Find the NE

Prompt AI with the matrix: **(D, D)** is the unique NE

Tool 4: Go Back to the Definition

Nash Equilibrium

A strategy profile is a NE if, **holding every other player's strategy fixed**, no single player can improve their payoff by unilaterally switching.

Apply to the candidate **(D, D)**:

1. Fix Player 2 at Defect.
   Player 1: Defect $= -2 >$ Cooperate $= -3$. ✓
2. Fix Player 1 at Defect.
   By symmetry, Player 2 has no incentive to deviate. ✓

Extension: NE to BNE

Example from before:

Pat privately knows her type (like or hate); Chris does not.

Each type has prob $\frac{1}{2}$ (common knowledge).

Pat(like) \ Chris	B	S
B	$(2,1)$	$(0,0)$
S	$(0,0)$	$(1,2)$

Pat(hate) \ Chris	B	S
B	$(0,1)$	$(2,0)$
S	$(1,0)$	$(0,2)$

Candidate BNE: Chris $= S$; Pat $= (S$ if like, B if hate$)$

1. Pat(like) given Chris$= S$: $EU(B) = 0$, $EU(S) = 1 \Rightarrow S$. ✓
2. Pat(hate) given Chris$= S$: $EU(B) = 2$, $EU(S) = 0 \Rightarrow B$. ✓
3. Chris given Pat$= (S, B)$: $EU(B) = \frac{1}{2}\cdot 0 + \frac{1}{2}\cdot 1 = \frac{1}{2}$, $EU(S) = \frac{1}{2}\cdot 2 + \frac{1}{2}\cdot 0 = 1 \Rightarrow S$. ✓

Extension: Procurement Auction

A buyer wants to procure one unit of a good.

- Example: Deicing Salt, MissouriBuys
- n suppliers (bidders); each has **private cost** $C_i \sim F(\cdot)$ i.i.d. on $[0, \bar{c}]$, with density $f > 0$.
  - F is common knowledge; c_i is bidder i's private **type**.

  (cf. **private value** $V_i \sim F(\cdot)$ on $[0, \bar{v}]$; **Uniform**$[0, 1]$ in both cases)
- Each bidder submits a bid $b_i \geq 0$ simultaneously.
- **Lowest bid wins; winner is paid her own bid.**
  (cf. **highest** bid wins; winner **pays** her own bid)
- Profit: $b_i - c_i$ if win, 0 if lose.
  (cf. $v_i - b_i$)

This is a Bayesian Game

- Players: $I = \{1, \ldots, n\}$.
- Types: costs $c_i \in [0, \bar{c}]$, i.i.d. from F.
- Actions: bids $b_i \in [0, \infty)$.
- Payoffs: $b_i - c_i$ if $b_i = \min_j b_j$, otherwise 0.

A pure strategy is a function $\beta_i : [0, \bar{c}] \to [0, \infty)$ mapping each cost type to a bid.

We look for a **symmetric** BNE: every bidder uses the same strictly *increasing* $\beta(\cdot)$.

[Full derivation in Appendix $\to$] [Another example: Cournot $\to$]

Prompt 1: Finding the Equilibrium

Copy into your AI session (start a new chat)

```
Consider a first-price sealed-bid procurement auction with n
     identical suppliers. Each supplier i privately observes
     cost c_i, drawn i.i.d. from CDF F(c) on [0, c_bar] with
     density f(c). Lowest bid wins; winner is paid her own
    bid. Winning profit: b_i - c_i.

Derive the symmetric pure-strategy BNE bidding strategy beta
    (c).


Show all steps.
```

What AI Returns: The Equilibrium

$$\beta(c) = \mathbb{E}[Y \mid Y > c] = c + \frac{\int_c^{\bar{c}} \left(1 - F(y)\right)^{n-1} dy}{\left(1 - F(c)\right)^{n-1}},$$

where $Y = \min_{j \neq i} C_j$ is the lowest rival cost.

For $n = 2$, Uniform$[0, 1]$ (procurement vs. lecture 3 buyer's auction):

	Procurement	Buyer's auction
Equilibrium	$\beta(c) = \frac{c+1}{2}$	$\beta(v) = \frac{v}{2}$
Direction	bid *above* cost	bid *below* value
Y	min rival cost	max rival value
Condition	$Y > c$ (win by low bid)	$Y < v$ (win by high bid)

Prompt 2: Economic Intuition (HW)

Continue in the same chat (follow-up to Prompt 1)

```
In the first-price procurement auction we just solved, the
    symmetric BNE is beta(c) = E[Y | Y > c] > c, so every
    supplier bids ABOVE their true cost. Please explain the
    economic intuition:
1. What trade-off does a supplier face when choosing their
    bid? Why does bidding lower help win more often but hurt
     profit per win?
2. Compare with the standard first-price buyer's auction (
    bidders shade BELOW their value). In what sense is
    procurement the exact flip side?
3. Write out the markup formula and explain each term.
4. What happens to the markup as n grows? Why does
    competition drive asking prices toward true costs?
```

Economic Intuition: Bidding Above Cost

$$\beta(c) = c + \underbrace{\frac{\int_c^{\bar{c}} \left(1 - F(y)\right)^{n-1} dy}{\left(1 - F(c)\right)^{n-1}}}_{\text{markup} > 0}$$

- **Trade-off:** lower bid $\Rightarrow$ win more often, earn less per win. Markup is the optimal balance.
- **Same math, opposite signs:** in Lecture 3 (buyer's auction), bidders shade *below* value; here suppliers shade *above* cost
- **Competition:** markup $\to 0$ as $n \to \infty \Rightarrow$ more suppliers drives bids toward true costs.

Tool 4: BNE Deviation Check

Setup: $n = 2$, $C \sim \text{Uniform}[0, 1]$, fix cost type $c = 0.3$

Candidate: $\beta(c) = \frac{c+1}{2}$, so known answer to verify: $\beta(0.3) = 0.65$.

Question: Does $\tilde{b} = 0.65$ yield the highest expected profit for cost type $c = 0.3$, given that all rivals follow $\beta(\cdot)$?

Tool 4: Simulation Overview

Fix every rival's strategy at the candidate $\beta(\cdot)$. *Simulate* expected profit of a bidder with $c = 0.3$ for *every* trial bid $\tilde{b}$ by simulating rivals. We want to verify if $\tilde{b} = 0.65$ yields the highest profit.

1. Draw $S = 1000$ rival costs $C^{(s)} \sim \text{Uniform}[0, 1]$.
   approximates $\mathbb{E}$ over rival types
2. Rival bids: $B^{(s)} = \beta(C^{(s)}) = \frac{C^{(s)}+1}{2}$.
3. For each trial bid $\tilde{b}$: $\widehat{\pi}(\tilde{b} \mid 0.3) = \frac{1}{S} \sum_s \mathbf{1}\{\tilde{b} < B^{(s)}\}(\tilde{b} - 0.3)$.
4. Argmax of $\widehat{\pi}(\cdot \mid 0.3)$ should equal $\beta(0.3) = 0.65$.
5. Repeat for a grid of c's; overlay $\hat{\beta}(c)$ and $\beta(c)$.
   BNE must hold for **every** type $c \in [0, \bar{c}]$!

Prompt 3: Ask AI to Write the Verification Code

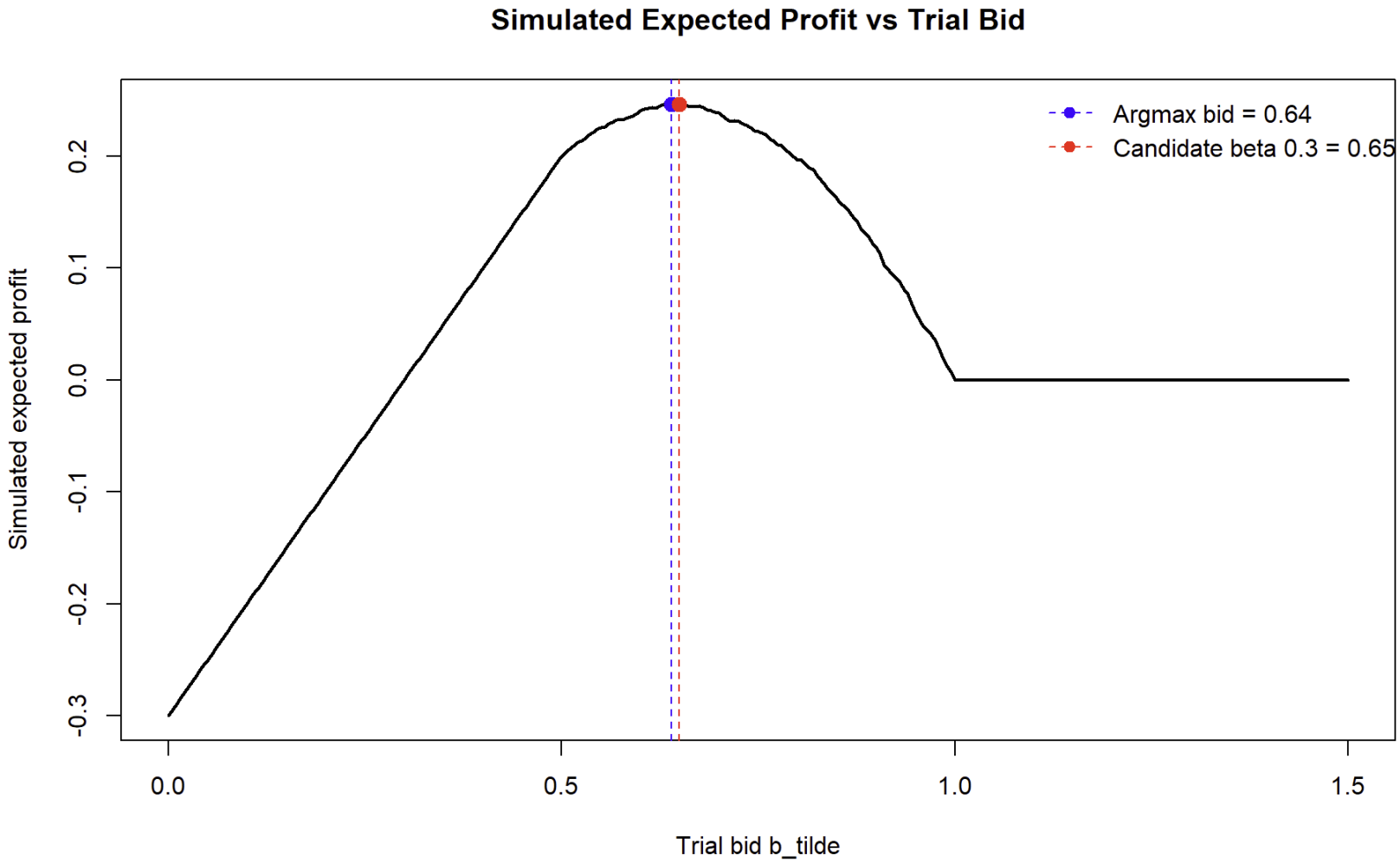
Continue in the same chat (follow-up to Prompt 1)

```
Write R code that verifies whether beta(c) = (c+1)/2 is
    truly the best-response bid for a supplier with cost c =
     0.3, when n = 2 and costs are Uniform[0,1]. The code
    must:

First Step. Define beta(c) = (c+1)/2 as the candidate
    strategy.
``
Explain to AI what you want it to do. You want AI to plot
    the expected profit against each trial bid.
''
Last Step. Print the profit-maximizing trial bid and compare
     with the candidate 0.65. If the two are equal, your
    candidate bid is indeed the equilibrium bid.

Do NOT compute expected profit analytically and use only the
     S simulated rival draws.
```

Simulation Result Example



Cautionary Note: Be Specific

The AI Already Knows the Answer

Vague instructions let AI satisfy you without truly testing the claim — e.g. quietly integrating analytically instead of simulating, or using a grid too coarse to reveal a deviation.

Specify, at minimum, for the simulation to work:

- Rival costs are *drawn randomly* (S explicit draws from F), not integrated analytically
- The exact profit rule (*payoffs*): win $\Rightarrow b - c$, lose $\Rightarrow 0$
- The bid grid (start, end, step size)
- The argmax should be read from the simulated numbers
- Profit curve must be plotted with the argmax overlaid on $\beta(c)$ (what is being tested against)

Computing an Expectation: Monte Carlo

Draw many values and average

$$\mathbb{E}[g(Y)] \approx \frac{1}{S}\sum_{s=1}^{S} g\big(Y^{(s)}\big), \qquad Y^{(s)} \sim \text{distribution of } Y.$$

The best-response check does exactly this: the average over S rival draws stands in for the expectation over rivals — which is why you never need a density.

How you draw depends on what you have:

- **A known distribution (this homework).** Built-in sampler (`np.random` / `runif`, `rnorm`). For a mixture (the two-hump): first flip a coin for the hump (`rbinom`$(1, 1, 0.5)$), then draw from that Normal.
- **Only a CDF F.** *Inverse-transform*: draw $U \sim \mathrm{Uniform}(0, 1)$, set $X = F^{-1}(U)$; then X has CDF F.
- **Only a data sample (no formula).** *Bootstrap*: resample the observed values with replacement — exactly drawing from the empirical CDF $\hat{F} =$ `my_ecdf`.

Random Draws or an Evenly Spaced Grid?

Two tools for two different jobs:

- **Averaging over uncertainty $\Rightarrow$ draw randomly.** When a quantity is genuinely random and you want its expectation, sample it. *Example:* the rival's cost type (you integrate profit over their distribution).
- **Sweeping a domain or a decision $\Rightarrow$ use a grid.** Searching over a choice (the trial bids $\tilde{b}$ in the argmax), evaluating/plotting a function (the cost types c where you report $\beta(c)$), or numerical integration (the y-grid inside the markup integral): use evenly spaced points. A grid gives even, reproducible coverage and, for a smooth 1-D integral, more accuracy per point than random draws.

Rule of thumb

Integrate over *randomness* $\rightarrow$ random draws; sweep a *domain* or a *decision variable* $\rightarrow$ a grid. This assignment needs **both**, used in the right places.

Homework: Verify β with Simulation

Use AI to write a code that verifies $\beta(c) = \frac{c+1}{2}$ is the symmetric BNE for $n = 2$, $C \sim \mathrm{Uniform}[0, 1]$, by computing the simulated best response for a grid of **ALL** c value types and plotting $\hat{\beta}(c)$ against $\beta(c)$.

Reuse Helper Functions — Don't Re-Derive Them

Write once, read many

You built `my_ecdf` and `my_kde` in HW1. Save them in *one* file and **load** them — do not ask AI to regenerate them in every chat.

- **Saves tokens & time:** re-generating (or re-pasting) the same function every session burns context you could spend on the actual problem.
- **Reproducible:** one canonical version ⇒ every question uses the *same* estimator, with no silent drift between chats.
- **Fewer bugs:** a function you already tested will not be quietly re-broken by a fresh (slightly different) AI rewrite.

R: keep helpers in one place, then just load them

```r
# helpers.R  <-- write my_ecdf, my_kde, trapz ONCE, here
source("helpers.R")    # load them at the top of every script
```

Python: put them in `helpers.py`, then `from helpers import my_ecdf, my_kde`.

You Still Need to Know Good Practice

AI implements; you decide *what* to implement

Having AI write the code does not remove the need to know **traditional good practice** — it makes it more valuable. You supply the standards; AI does the typing.

- **Seed the RNG** (`set.seed` / `default_rng`) so results reproduce — ask AI to do it, but *you* have to know to ask.
- **Modularize:** small, named, reusable functions (your helpers) instead of one long copy-pasted block.
- **Sanity-check against a known answer** before trusting the general case — exactly the Tool 4 / uniform $\frac{c+1}{2}$ check.
- **Comment and name** variables so you — and a grader — can read it.

Which Equilibrium?

There Can Be More Than One

Confirming a candidate survives the deviation test shows it *is a* BNE, but not if it is the *only* one.

- Asymmetric pure-strategy BNE exist even among ex ante symmetric bidders (Maskin & Riley, 2000).
- An alternative equilibrium can generate a bid distribution that is **observationally indistinguishable** from the symmetric one under some cost distribution

$\rightarrow$ Bid data alone cannot tell you which equilibrium is played.

- We focus on the symmetric, strictly increasing, pure-strategy BNE

References

Maskin, E., & Riley, J. (2000). Asymmetric auctions. *The Review of Economic Studies*, **67**(3), 413–438.

Appendix: Cournot Duopoly

Two firms choose quantities simultaneously. Inverse demand $P = a - q_1 - q_2$, marginal cost m:

$$\pi_i(q_i;\, q_j) = (a - q_i - q_j - m)\, q_i.$$

1. **FOC in own choice, rival held fixed:**
   $a - 2q_i - q_j - m = 0 \;\Rightarrow\; BR_i(q_j) = \dfrac{a - m - q_j}{2}.$
2. **Then impose equilibrium (symmetry)** $q_i = q_j = q^*$:
   $q^* = \dfrac{a - m - q^*}{2} \;\Rightarrow\; q^* = \dfrac{a - m}{3}.$
3. **Tool 4:** fix $q_j = \frac{a-m}{3}$; then $BR_i\left(\frac{a-m}{3}\right) = \frac{a-m}{3} = q^*$. ✓

Order Matters

Impose $q_i = q_j = q$ *before* the FOC: $\max_q (a - 2q - m)q \Rightarrow q = \frac{a-m}{4}$ — the cartel output, **not** a NE. The firm no longer holds its rival fixed. Same reason we impose $z = c$ only *after* the FOC.

[← Back to main] [Next: Cournot vs Auction →]

Appendix: Cournot vs Auction

	Cournot	Procurement auction
Own choice	q_i (a number)	bid b (a number)
Held fixed	rival's q_j	rivals' strategy $\beta(\cdot)$; own type c
FOC	$\partial \pi_i / \partial q_i = 0$	$\partial \Pi / \partial b = 0$
Equilibrium step	$q_i = q_j = q^*$	$b = \beta(c)$
Best response maps	number $\to$ number	function $\beta(\cdot) \to$ function of c
Equilibrium	$q^* = BR(q^*)$	$\beta = BR(\beta)$
Rival's cost	known: m (common)	private: $c_j \sim F$ (uncertain)
Solution concept	**NE** (complete info)	**BNE** (private types)

$\beta(c)$ Is Not a Best-Response Curve

$\beta(c)$ is the **equilibrium strategy** — the analog of q^*, not of $BR(q_j)$. The best response takes rivals' whole function $\beta(\cdot)$ and returns the optimal bid for each c: that is $\hat{\beta}(c)$ in the simulation. Checking $\hat{\beta} = \beta$ is checking the fixed point.

[← Back to main] [Next: Auction derivation (bid space) →]

Appendix: Solving the Procurement Auction — Setup

Guess: β is strictly increasing. Given cost c, instead of choosing bid b directly, choose z such that $\beta(z) = b$ ("pretend to have cost z"):

$$\Pi(z;\, c) = \underbrace{(1 - F(z))^{n-1}}_{=:\, H(z),\ \Pr(\text{win})} \cdot (\beta(z) - c).$$

c is **fixed**; z is the **variable you choose to optimize**. This distinction is where AI most often goes wrong (see main slides). Take the FOC with respect to z *first*; impose $z = c$ only *after*.

[← Back to main] [Next: bid-space route →]

Appendix: The Same Problem in Bid Space (matches the simulation)

The Setup chose z with $\beta(z) = b$. You can instead choose the **bid** b **directly** — the more natural formulation, and exactly what the Tool-4 simulation sweeps. It gives the same ODE.

Two things to keep straight

1. $H(\cdot)$ **lives in cost space.** $H(x) = (1 - F(x))^{n-1}$ takes a *cost* as its input — never a bid.
2. $\beta(\cdot)$ **is the rivals' bid function**, held fixed. You choose one number b; β maps a *rival's* cost to that rival's bid.

A bid b wins iff it undercuts every rival's bid:

$$b < \beta(C_j) \;\; \forall j \iff C_j > \beta^{-1}(b) \;\; \forall j \implies \Pr(\text{win}) = H\big(\beta^{-1}(b)\big).$$

$$\Pi(b;\, c) = H\big(\beta^{-1}(b)\big)\,(b - c).$$

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[Next: FOC in b →]

Appendix: Bid-Space FOC $\to$ the Same ODE

Take the FOC with respect to the **number** b (the bidder chooses one bid; β is the rivals' *fixed* strategy):

$$\frac{H'\big(\beta^{-1}(b)\big)}{\beta'\big(\beta^{-1}(b)\big)}\,(b-c)+H\big(\beta^{-1}(b)\big)=0.$$

Impose the equilibrium requirement $b=\beta(c)$, so $\beta^{-1}(b)=c$:

$$\frac{H'(c)}{\beta'(c)}\big(\beta(c)-c\big)+H(c)=0.$$

Multiply by $\beta'(c)$:

$$H'(c)\big(\beta(c)-c\big)+H(c)\,\beta'(c)=0,$$

exactly the ODE from the z-approach.

Why the slides use z

The substitution $b=\beta(z)$ removes β^{-1}. By the chain rule $\partial\Pi/\partial z=(\partial\Pi/\partial b)\,\beta'(z)$, and $\beta'>0$, so one FOC vanishes exactly when the other does. "Choose z" and "choose b" are the same problem in different units.

Appendix: Uniform Check in Bid Space

For $n = 2$, $C \sim \text{Uniform}[0, 1]$: $\beta(c) = \frac{c+1}{2}$, so $\beta^{-1}(b) = 2b - 1$ and $H(x) = 1 - x$. Then

$$\Pr(\text{win}) = H(2b-1) = 1-(2b-1) = 2(1-b), \qquad \Pi(b;\, c) = 2(1-b)(b-c$$

FOC in b:

$$-(b - c) + (1 - b) = 0 \implies b = \frac{1 + c}{2} = \beta(c).$$

The best bid drops straight out of the FOC — **nothing imposed**. This is the verification in its most transparent form, and it is exactly what the Tool-4 simulation recovers by sweeping $\tilde{b}$.

[← Back to main] [Next: the z-derivation →]

Appendix: First-Order Condition

$$\left.\frac{\partial \Pi}{\partial z}\right|_{z=c} = H'(c)\big(\beta(c) - c\big) + H(c)\,\beta'(c) = 0.$$

Because β is a BNE strategy, the true type c must be the optimizer, so the FOC holds at $z = c$. Rearranging:

$$H(c)\,\beta'(c) + H'(c)\,\beta(c) = c\,H'(c).$$

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[Next: ODE →]

Appendix: Solving the ODE

The left-hand side is $\frac{d}{dc}[H(c)\,\beta(c)]$:

$$\frac{d}{dc}\big[H(c)\,\beta(c)\big] = c\,H'(c).$$

Integrate from c to $\bar{c}$, using boundary condition $H(\bar{c}) = 0$ (the highest-cost type wins with probability zero and is therefore indifferent; $\beta(\bar{c}) = \bar{c}$):

$$H(c)\,\beta(c) = \int_c^{\bar{c}} y\,\big(-H'(y)\big)\,dy.$$

[← Back to main]

[Next: Solution →]

Appendix: The BNE Bidding Strategy

$$\beta(c) = \frac{1}{H(c)} \int_c^{\bar{c}} y\,(n-1)(1-F(y))^{n-2} f(y)\,dy$$
$$= \mathbb{E}[Y \mid Y > c], \qquad Y := \min_{j \neq i} C_j.$$

Markup Form (integration by parts)

$$\beta(c) = c + \frac{\int_c^{\bar{c}} (1-F(y))^{n-1}\,dy}{(1-F(c))^{n-1}}.$$

Bidders ask *above* their true cost; markup shrinks as $n \to \infty$.

[← Back to main] [Next: Verification →]

Appendix: Verification (Optional)

Using $\frac{d}{dz}[H(z)\beta(z)] = zH'(z)$:

$$\frac{\partial \Pi}{\partial z} = zH'(z) - cH'(z) = H'(z)(z - c).$$

Since $H' < 0$: $\partial\Pi/\partial z > 0$ for $z < c$ and < 0 for $z > c$. The profit is **single-peaked** at $z = c$.
$\Rightarrow \beta$ is indeed bidder i's best reply when all rivals use $\beta \Rightarrow$ symmetric BNE confirmed.

[$\leftarrow$ Back to main] [Next: Buyer's Payment $\rightarrow$]

Appendix: Buyer's Expected Payment

The winner is the lowest-cost supplier $C_{(n)} := \min_i C_i$, paid

$$\beta(C_{(n)}) = \mathbb{E}[C_{(n-1)} \mid C_{(n)}],$$

where $C_{(n-1)}$ is the second-lowest cost.
Taking expectations:

$$\mathbb{E}\big[\beta(C_{(n)})\big] = \mathbb{E}[C_{(n-1)}].$$

The buyer pays the expected second-lowest cost — an instance of **Revenue Equivalence**: the second-price procurement auction (lowest bidder wins, paid the second-lowest bid) yields the same expected payment.

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